

# Case File

## James R. Whitfield

Liver transplant admission, September 22 to 29, 2026

Chart as it stood at 12:50 on post-operative day 6, twelve minutes before the alert storm

**61-year-old man**, six days after a deceased-donor liver transplant, with chronic kidney disease, type 2 diabetes, atrial fibrillation, and 22 active medication orders.

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*Fictional chart for a teaching simulation. The patient, staff, hospital, and all values are invented. Not for clinical use.*

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Face sheet

Mr. Whitfield is 6 days out from a liver transplant, with kidney disease, diabetes, atrial fibrillation, and 22 active drug orders that interact with each other. His anti-rejection drug has a narrow safe range, so his labs, doses, and orders change every day and must be checked several times a shift.

| Field                 | Value                                                             |
|-----------------------|-------------------------------------------------------------------|
| Patient               | WHITFIELD, James R.                                               |
| Age, sex              | 61, male                                                          |
| Medical record number | RUH-4471902                                                       |
| Hospital              | Riverbend University Hospital                                     |
| Unit, bed             | 6 West, transplant step-down, bed 14                              |
| Admitted              | September 22, 2026, 21:40                                         |
| Procedure             | Deceased-donor liver transplant, September 23, 2026               |
| Chart date            | September 29, 2026 (post-op day 6)                                |
| Attending             | Priya Raman, MD, transplant hepatology                            |
| Surgeon               | Thomas Okafor, MD, transplant surgery                             |
| Resident              | Liam Chen, MD                                                     |
| Nurse, day shift      | Maya Torres, RN                                                   |
| Pharmacist            | Dana Patel, PharmD                                                |
| Case manager          | Rosa Alvarez, RN                                                  |
| Code status           | Full code                                                         |
| Health care proxy     | Emily Whitfield, daughter                                         |
| Height, weight        | 178 cm; 94.2 kg today (88 kg dry weight)                          |
| Blood type            | O positive                                                        |
| Isolation             | Protective precautions, immunosuppressed                          |
| Insurance             | Commercial plan, prior authorization required for specialty drugs |

Allergies

| Substance  | Reaction        | Source                                   | Verified                                                |
|------------|-----------------|------------------------------------------|---------------------------------------------------------|
| Penicillin | Rash, childhood | Patient report, 2011                     | No. Tolerated cefazolin during surgery on September 23. |
| Lisinopril | High potassium  | Outside clinic note, not in allergy list | No                                                      |

Active problem list (as shown in the EHR)

- 1
- Status post liver transplant, post-op day 6
- 2
- Cirrhosis due to fatty liver disease (MASH)
- 3
- Ascites

- 4 Hepatic encephalopathy, grade 2
- 5 Type 2 diabetes with steroid-induced high glucose
- 6 Chronic kidney disease, stage 3b
- 7 Paroxysmal atrial fibrillation
- 8 Hypertension
- 9 Major depressive disorder, in remission
- 10 Low platelets
- 11 Fever, new today

Problems 2, 3, and 4 describe the liver that was removed. They have not been closed out.

## Critical status at 12:45, September 29

Mr. Whitfield is getting worse, and eight separate dangers are open at once. His early warning score has climbed from 0 to 5 in 8 hours. A score of 5 or more calls for urgent physician review, and none has been requested.

### Early warning score, September 29

| Time  | Score | What changed                                             |
|-------|-------|----------------------------------------------------------|
| 04:00 | 0     | Stable                                                   |
| 08:00 | 1     | Heart rate 92                                            |
| 10:30 | 3     | Temperature 38.1 C, heart rate 104, oxygen 95%           |
| 12:00 | 5     | Oxygen 93% and now on 2 liters of oxygen, heart rate 108 |

**Sepsis prediction model:** risk 0.71, labeled "high," first fired at 10:42. The chart shows no accuracy figures, no owner, and no guidance for transplant patients on steroids.

### Open dangers

| # | Danger                                       | Evidence in the chart                                                                                                                 | Where the evidence sits                          |
|---|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| 1 | Tacrolimus toxicity                          | Level 11.6 and rising; two interacting drugs added; hand tremor at 05:30                                                              | Lab tab, three consult notes, night nursing note |
| 2 | Acute kidney injury                          | Urine 26 mL per hour; weight up 1.3 kg; creatinine pending                                                                            | Intake and output screen                         |
| 3 | Dangerous potassium                          | Rising 3 days, 5.0 yesterday; emergency visit for potassium 6.4 on September 3                                                        | Lab tab; outside fax                             |
| 4 | Infection the antibiotic may not cover       | Fever; donor blood culture grew a resistant bacterium; his own resistant infection in March                                           | Transplant module; outside fax                   |
| 5 | Reduced blood flow in the new liver's artery | Ultrasound flow reading below the expected range                                                                                      | Body of the radiology report, not its summary    |
| 6 | Breathing                                    | New oxygen need; fluid around the right lung; untreated sleep apnea; on opioids                                                       | Nursing flowsheet; chest x-ray; outside fax      |
| 7 | Clot versus bleed                            | Blood thinner on hold with atrial fibrillation, a heart stent, and a past portal vein clot; returned to surgery for bleeding on day 1 | Cardiology note; outside fax                     |
| 8 | Heart rhythm                                 | QTc 492 ms on four QT-lengthening drugs; magnesium 1.4 and falling                                                                    | ECG tab; lab tab                                 |

No single screen shows these eight items together.

## Transplant admission history and physical

**Date:** September 22, 2026, 22:15. **Author:** Liam Chen, MD. **Cosigned:** Priya Raman, MD.

**Reason for admission:** Called in for a deceased-donor liver offer.

### History of present illness

Mr. Whitfield is a 61-year-old man with cirrhosis from fatty liver disease, diagnosed in 2021. He was listed for transplant in February 2026. His MELD-Na score at admission is 32.

Over the past 12 months he had 4 hospital admissions at Carver County Community Hospital. Two were for confusion from hepatic encephalopathy. One was for infected abdominal fluid in March 2026. One was for acute kidney injury after a large-volume fluid drainage in June 2026.

He has needed abdominal fluid drained every 2 to 3 weeks since April. The last drainage was September 14, with 6.5 liters removed. He reports fatigue, poor appetite, and a 5 kg loss of muscle weight over 6 months. He denies fever, black stools, vomiting blood, or chest pain. His daughter reports mild confusion in the evenings.

### Past medical history

| Condition                        | Since         | Details                                                                                                                    |
|----------------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------|
| Cirrhosis, fatty liver disease   | 2021          | Ascites, encephalopathy, small esophageal varices banded in 2024                                                           |
| Type 2 diabetes                  | 2012          | Metformin stopped in 2025 for kidney function. On insulin since. Last HbA1c 7.9% in August 2026.                           |
| Chronic kidney disease, stage 3b | 2023          | Baseline creatinine 1.4 to 1.6. Followed by Dr. Alan Brooks, Lakeshore Kidney Associates. Records requested, not received. |
| Paroxysmal atrial fibrillation   | 2022          | On blood thinner and diltiazem. CHA2DS2-VASc score 2.                                                                      |
| Hypertension                     | 2008          | Lisinopril stopped for high potassium, date unknown                                                                        |
| Major depressive disorder        | 2019          | Stable on sertraline                                                                                                       |
| Obesity                          | Long-standing | BMI 34 in 2021, now 28 at dry weight                                                                                       |
| Knee osteoarthritis              | 2017          | Uses over-the-counter pain relievers                                                                                       |

**Past surgical history:** Gallbladder removal in 2009. Right knee arthroscopy in 2017. Variceal banding in 2024.

### Home medications: three sources, three different lists

| Drug           | List A: outside hospital discharge, March 2026 | List B: transplant clinic, August 2026 | List C: daughter by phone, September 29    |
|----------------|------------------------------------------------|----------------------------------------|--------------------------------------------|
| Blood thinner  | Warfarin 5 mg daily                            | Apixaban 5 mg twice daily              | Apixaban 5 mg twice daily                  |
| Diabetes       | Metformin 1000 mg twice daily                  | Insulin glargine 14 units nightly      | Insulin glargine 14 units nightly          |
| Furosemide     | 40 mg daily                                    | 40 mg daily                            | 40 mg daily                                |
| Spironolactone | 100 mg daily                                   | 50 mg daily                            | Stopped early September for high potassium |
| Lactulose      | 30 mL three times daily                        | 30 mL three times daily                | Taking                                     |
| Rifaximin      | 550 mg twice daily                             | 550 mg twice daily                     | Taking                                     |
| Sertraline     | 50 mg daily                                    | 100 mg daily                           | 100 mg daily                               |

| Drug           | List A: outside hospital discharge, March 2026 | List B: transplant clinic, August 2026 | List C: daughter by phone, September 29   |
|----------------|------------------------------------------------|----------------------------------------|-------------------------------------------|
| Blood pressure | Lisinopril 10 mg daily                         | Amlodipine 5 mg daily                  | Amlodipine 5 mg daily                     |
| Diltiazem ER   | Not listed                                     | 120 mg daily                           | 120 mg daily                              |
| Pantoprazole   | Not listed                                     | 40 mg daily                            | 40 mg daily                               |
| Ibuprofen      | Not listed                                     | Not listed                             | 400 mg "a few times a week" for knee pain |

The EHR imported List A at admission. List B is in a scanned clinic letter. List C was taken by phone on post-op day 6 and written in a pharmacy note.

**Family history:** Father died of a heart attack at 67. Mother had type 2 diabetes. One brother with fatty liver disease.

**Social history:** Retired school bus mechanic. Widowed in 2020. Lives alone in Carver County, 90 miles from the hospital. Daughter Emily lives 20 minutes from him and will be his caregiver. Former smoker, 20 pack-years, quit 2009. No alcohol since 2021. No drug use.

#### Examination at admission

- Vitals: temperature 36.8 C, heart rate 84 irregular, blood pressure 108/62, oxygen saturation 96% on room air
- General: thin arms and legs, yellow skin and eyes
- Mental status: oriented to person and place, slow to answer, mild hand flapping
- Abdomen: distended with fluid, not tender
- Legs: swelling to the knees

#### Laboratory results at admission

| Test            | Result                                   | Normal range  |
|-----------------|------------------------------------------|---------------|
| Total bilirubin | 9.4 mg/dL                                | 0.2 to 1.2    |
| INR             | 2.3                                      | 0.9 to 1.1    |
| Creatinine      | 1.6 mg/dL                                | 0.7 to 1.3    |
| Sodium          | 129 mmol/L                               | 136 to 145    |
| Potassium       | 4.9 mmol/L                               | 3.5 to 5.0    |
| Albumin         | 2.4 g/dL                                 | 3.5 to 5.0    |
| Platelets       | 48 thousand/uL                           | 150 to 400    |
| Hemoglobin      | 9.1 g/dL                                 | 13.5 to 17.5  |
| Glucose         | 212 mg/dL                                | 70 to 140     |
| ECG             | Atrial fibrillation, rate 84, QTc 452 ms | QTc under 450 |

**Viral status:** Donor positive for cytomegalovirus, recipient negative. This is the highest-risk pairing and requires 6 months of antiviral prevention.

**Assessment and plan:** End-stage liver disease, MELD-Na 32, accepted for transplant. Hold apixaban. Consent signed. To operating room when donor team confirms.

## Outside records

An 84-page fax from Carver County Community Hospital and Lakeshore Kidney Associates arrived on September 29 at 10:05. It contains nine facts that change today's decisions. It was scanned as one image file named "Outside records," cannot be searched, and had not been opened by 12:45.

| Date         | Event in outside records                                                                                        | Why it matters today                                                 |
|--------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Oct 2023     | Heart stent placed after chest pain. Told to take aspirin for life.                                             | Affects the blood thinner decision. Not in the problem list.         |
| Nov 2023     | Kidney injury after CT contrast dye. Creatinine rose from 1.3 to 2.2.                                           | Radiology now suggests a CT scan with contrast for the liver artery. |
| Mar 2025     | 2.1 cm liver cancer treated with heat ablation. Scans clear since.                                              | Known to the transplant clinic. Missing from the admission note.     |
| Mar 2026     | Infected abdominal fluid grew <i>E. coli</i> resistant to penicillins and cephalosporins. Cured with meropenem. | Today's antibiotic is a penicillin-type drug.                        |
| Jun 2026     | Kidney failure after fluid drainage. Creatinine peaked at 3.1. Needed 5 days of IV treatment.                   | His kidneys have little reserve.                                     |
| Jul 2026     | Partial clot in the portal vein on CT.                                                                          | A reason to restart the blood thinner sooner.                        |
| Aug 2026     | Sleep study: moderate sleep apnea. Breathing machine prescribed, not used.                                      | He is on opioids and now needs oxygen.                               |
| Sep 3, 2026  | Emergency visit. Potassium 6.4 with ECG changes while on spironolactone.                                        | He reached a dangerous potassium 26 days ago.                        |
| 2024 to 2026 | Kidney clinic: creatinine 1.3 in 2024, 1.6 in 2026. Protein in urine.                                           | His true baseline is lower than the chart assumes.                   |

**Also in the fax:** a note that he missed 2 of his last 6 clinic visits for lack of a ride, and a pharmacy record showing ibuprofen 600 mg filled monthly since 2024.

**What the transplant team's own records hold.** The cancer, the stent, and the portal vein clot are recorded in the transplant evaluation module. That module is a separate program. Its contents do not appear in the main chart's problem list, allergy list, or admission note.

## Donor organ and pathology

The donor liver came with higher than average risk, and a donor blood culture grew bacteria that today's antibiotic does not cover. That culture result reached the transplant coordinator on September 26 and was never entered in the main chart.

### Donor and organ

| Item                                   | Detail                                                                                       |
|----------------------------------------|----------------------------------------------------------------------------------------------|
| Donor                                  | 54 years old, donation after circulatory death                                               |
| Time without blood flow before cooling | 19 minutes                                                                                   |
| Fat in donor liver on biopsy           | 20%                                                                                          |
| Virus status                           | Cytomegalovirus positive                                                                     |
| Known risks of this organ type         | Bile duct injury, slow early function, artery problems                                       |
| Donor blood culture                    | <i>Klebsiella pneumoniae</i> . Preliminary result phoned to coordinator September 26, 16:10. |
| Final sensitivities, September 28      | Resistant to piperacillin-tazobactam. Sensitive to meropenem.                                |
| Where recorded                         | Transplant module, donor tab. No order, alert, or note in the main chart.                    |

**Pathology of the removed liver.** Reported September 27. Routed to the surgeon's inbox, unread.

- Cirrhosis with severe scarring.
- The treated 2.1 cm tumor shows no living cancer.
- A second, untreated cancer of 0.9 cm was found. It was not seen on any scan.
- No spread into blood vessels. Edges clear.
- Recommendation: cancer surveillance scans every 3 months. Discuss lower-intensity immune suppression at 3 months.

Neither the donor culture nor the second tumor is mentioned in any physician note.

## Operative summary and hospital course

**Operative summary, September 23, 2026.** Surgeon: Thomas Okafor, MD.

| Item                             | Detail                                               |
|----------------------------------|------------------------------------------------------|
| Procedure                        | Whole-liver transplant from a deceased donor         |
| Bile duct connection             | Duct to duct, no stent                               |
| Operative time                   | 7 hours 10 minutes                                   |
| Cold storage time of donor liver | 6 hours 40 minutes                                   |
| Estimated blood loss             | 2.8 liters                                           |
| Transfusions                     | 6 units red cells, 4 units plasma, 2 units platelets |
| Fluid removed from abdomen       | 5.2 liters                                           |
| Drains                           | 2 surgical drains, right upper abdomen               |
| Antibiotic at incision           | Cefazolin, tolerated without reaction                |
| Immune suppression at surgery    | Methylprednisolone 500 mg and basiliximab 20 mg      |
| Complications                    | Blood pressure collapse for 4 minutes at reperfusion |

Basiliximab was chosen so that tacrolimus could be started late and low, to protect his kidneys.

### Hospital course by day

| Day   | Date   | Events                                                                                                                                                                                                                                                                                                    |
|-------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Day 0 | Sep 23 | Transplant. Blood pressure collapsed for 4 minutes when the new liver was connected, treated with epinephrine. To intensive care on a ventilator and two blood pressure drugs.                                                                                                                            |
| Day 1 | Sep 24 | Drain put out 900 mL of blood in 3 hours overnight. Returned to the operating room at 02:20. 1.2 liters of clot removed and one bleeding vessel tied. 4 more units of red cells. Ultrasound afterward: artery and veins open. Tacrolimus started at 2 mg twice daily. Insulin drip for glucose above 300. |
| Day 2 | Sep 25 | Atrial fibrillation with heart rate 150 at 03:10. Diltiazem drip started. Breathing tube removed at 10:15. Urine output low, creatinine 1.9, kidney team consulted. Confused and pulling at lines overnight. One dose of haloperidol given.                                                               |
| Day 3 | Sep 26 | Off blood pressure drugs. Confusion cleared. Changed to oral diltiazem ER 120 mg. Insulin drip changed to injections. Tacrolimus raised to 3 mg twice daily. Cloudy drain fluid sent for culture. Cardiology consulted. Donor culture result phoned to coordinator.                                       |
| Day 4 | Sep 27 | Yeast grew from drain fluid. Fluconazole 200 mg daily started. Infectious disease consulted. Tacrolimus raised to 4 mg twice daily. Second basiliximab dose given. Transferred from intensive care to 6 West at 18:00.                                                                                    |
| Day 5 | Sep 28 | Liver numbers improving. Ultrasound: vessels open, artery flow reading low. Tacrolimus level 11.6, at the top of target. Dose left at 4 mg twice daily. Potassium 5.0. Prednisone taper to 20 mg. One drain removed. Refused evening mycophenolate for nausea.                                            |
| Day 6 | Sep 29 | Hand tremor and low urine output reported at 05:30. Temperature 38.1 C at 10:30. Cultures sent. Piperacillin-tazobactam started at 11:00. New oxygen need at 12:00. Chest x-ray: fluid around right lung. Morning labs delayed, resulted at 13:02.                                                        |

**Drug interaction timeline.** Diltiazem (day 3) and fluconazole (day 4) both slow the breakdown of tacrolimus. The tacrolimus dose was raised on both of those days. No dose reduction was ordered when either drug started.

## Lab trends, glucose, and vitals

The new liver is recovering well, but the tacrolimus level has risen every day and passed the target range on day 6. Potassium and creatinine rose with it.

### Daily morning labs, post-op days 1 to 6

| Test (target or normal)                  | Day 1     | Day 2 | Day 3 | Day 4 | Day 5 | Day 6 |
|------------------------------------------|-----------|-------|-------|-------|-------|-------|
| AST, U/L (under 40)                      | 842       | 510   | 290   | 164   | 98    | 76    |
| ALT, U/L (under 40)                      | 615       | 480   | 350   | 241   | 170   | 132   |
| Total bilirubin, mg/dL (under 1.2)       | 5.8       | 4.6   | 3.5   | 2.7   | 2.1   | 1.8   |
| INR (under 1.2)                          | 1.9       | 1.6   | 1.4   | 1.3   | 1.2   | 1.2   |
| Creatinine, mg/dL (baseline 1.4 to 1.6)  | 1.7       | 1.9   | 1.7   | 1.5   | 1.6   | 2.4   |
| Potassium, mmol/L (3.5 to 5.0)           | 4.4       | 4.3   | 4.5   | 4.7   | 5.0   | 6.2   |
| Magnesium, mg/dL (1.7 to 2.2)            | 1.8       | 1.6   | 1.5   | 1.5   | 1.4   | 1.3   |
| Tacrolimus level, ng/mL (target 8 to 12) | Not drawn | 3.2   | 5.1   | 7.4   | 11.6  | 19.4  |
| White cells, thousand/uL (4 to 11)       | 9.8       | 11.2  | 12.5  | 13.1  | 13.8  | 14.6  |
| Platelets, thousand/uL (150 to 400)      | 54        | 58    | 66    | 79    | 92    | 104   |
| Hemoglobin, g/dL (13.5 to 17.5)          | 8.4       | 8.1   | 8.3   | 8.6   | 8.8   | 8.7   |

Day 6 labs were drawn at 05:40, clotted, and were redrawn at 08:20. They resulted at 13:02.

### Glucose and insulin, last 24 hours

| Date, time    | Glucose, mg/dL | Insulin given                          |
|---------------|----------------|----------------------------------------|
| Sep 28, 12:00 | 301            | 8 units lispro                         |
| Sep 28, 17:00 | 284            | 6 units lispro                         |
| Sep 28, 21:00 | 244            | 4 units lispro, plus 18 units glargine |
| Sep 29, 02:00 | 198            | 2 units lispro                         |
| Sep 29, 06:00 | 247            | 4 units lispro                         |
| Sep 29, 08:00 | 268            | 6 units lispro                         |
| Sep 29, 12:30 | 312            | 8 units lispro due                     |

### Vital signs, September 29

| Time  | Temperature | Heart rate     | Blood pressure | Breathing rate | Oxygen                    |
|-------|-------------|----------------|----------------|----------------|---------------------------|
| 04:00 | 37.2 C      | 88             | 142/84         | 16             | 96%                       |
| 08:00 | 37.4 C      | 92             | 148/88         | 18             | 96%                       |
| 10:30 | 38.1 C      | 104            | 152/90         | 18             | 95%                       |
| 12:00 | 37.9 C      | 108, irregular | 156/92         | 20             | 93% on 2 liters of oxygen |

### Intake, output, and weight

| Measure               | Sep 28                | Sep 29 so far     |
|-----------------------|-----------------------|-------------------|
| Urine output          | 1,150 mL in 24 hours  | 210 mL in 8 hours |
| Surgical drain output | 180 mL, thin and pink | 60 mL in 8 hours  |
| Oral and IV intake    | 2,300 mL              | 900 mL in 8 hours |
| Weight                | 92.9 kg               | 94.2 kg           |

**Other results**

- Liver ultrasound, September 28: artery, portal vein, and hepatic veins open. No fluid collection. Report filed under scanned media.
- ECG, September 28: atrial fibrillation, rate 96, QTc 468 ms.
- Drain fluid culture, September 26: *Candida albicans*, light growth.
- Blood and urine cultures, September 29: pending.

## Imaging and test reports

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Three reports from the last 24 hours contain findings that are not in any physician note. In each case the important line sits in the body of the report and not in its summary.

**Liver ultrasound with blood flow study.** September 28, 15:10. Filed under scanned media.

- Summary line: "Vessels open. No fluid collection."
- Body of report: hepatic artery resistive index 0.48. Expected range is 0.55 to 0.80. On day 1 it was 0.66.
- Body of report: a falling reading can be an early sign of artery narrowing.
- Recommendation: repeat ultrasound in 24 hours, or CT scan with contrast if liver numbers rise.
- Small amount of fluid around the right lung.
- No repeat ultrasound has been ordered.

**Chest x-ray.** September 29, 11:15.

- Moderate fluid around the right lung, larger than on September 26.
- New haziness in the right lower lung. Could be collapsed lung tissue or early pneumonia.
- Central line tip in correct position.
- Result sent to the ordering resident's inbox at 11:52.

**ECG.** September 29, 11:40.

- Atrial fibrillation, rate 110.
- QTc 492 ms. It was 452 at admission and 468 yesterday.
- Machine reading: "Prolonged QT. Abnormal ECG." Not yet reviewed by a physician.

**Bedside lactate.** September 29, 11:45. Result 2.9 mmol/L, normal under 2.0. A raised lactate with fever suggests the body is under strain from infection or poor blood flow. Recorded in the nursing flowsheet only.

**Heart ultrasound before transplant.** August 2026. Pumping function normal at 55%. Left upper chamber mildly enlarged. Lung artery pressure mildly raised.

## Microbiology

He has had two bacteria resistant to today's antibiotic, one of his own and one from the donor. Both results sit outside the main chart.

| Date          | Sample                            | Result                                                                                | Where it is recorded                |
|---------------|-----------------------------------|---------------------------------------------------------------------------------------|-------------------------------------|
| Mar 2026      | Abdominal fluid, outside hospital | <i>E. coli</i> , resistant to penicillins and cephalosporins. Sensitive to meropenem. | Outside fax, page 31                |
| Sep 22        | Nose swab at admission            | MRSA carrier                                                                          | Lab tab. No isolation order placed. |
| Sep 22        | Donor blood                       | <i>Klebsiella pneumoniae</i> , resistant to piperacillin-tazobactam                   | Transplant module only              |
| Sep 26        | Surgical drain fluid              | <i>Candida albicans</i> , light growth                                                | Lab tab                             |
| Sep 28        | Blood test for cytomegalovirus    | Not detected                                                                          | Lab tab                             |
| Sep 29, 10:50 | Blood, 2 sites                    | Pending                                                                               | Lab tab                             |
| Sep 29, 10:50 | Urine                             | Pending                                                                               | Lab tab                             |

**Antibiotics given this admission:** cefazolin for 24 hours around surgery; fluconazole since September 27; piperacillin-tazobactam since 11:20 today.

**Not ordered:** stool test for *C. difficile* despite loose stools, drain fluid re-culture, and sputum culture.

## Active medication orders

Mr. Whitfield has 22 active orders on September 29, and most of them need a lab value or a vital sign checked before or after each dose. Two orders, lactulose and rifaximin, treat a condition he no longer has.

| #  | Drug and dose                                                   | Purpose                          | What staff must monitor                                                                                       |
|----|-----------------------------------------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------|
| 1  | Tacrolimus 4 mg by mouth, 09:00 and 21:00                       | Anti-rejection                   | Blood level every morning before the dose. Creatinine, potassium, magnesium, tremor, blood pressure, glucose. |
| 2  | Mycophenolate 1,000 mg twice daily                              | Anti-rejection                   | White cells, platelets, diarrhea                                                                              |
| 3  | Prednisone 20 mg daily, tapering                                | Anti-rejection                   | Glucose, blood pressure, mood, wound healing                                                                  |
| 4  | Valganciclovir 450 mg daily                                     | Prevents cytomegalovirus         | Dose depends on kidney function. White cells.                                                                 |
| 5  | Trimethoprim-sulfamethoxazole 80/400 mg daily                   | Prevents pneumonia               | Potassium, creatinine                                                                                         |
| 6  | Fluconazole 200 mg daily, started day 4                         | Yeast in drain fluid             | Raises tacrolimus level. QTc. Liver numbers.                                                                  |
| 7  | Insulin glargine 18 units at 21:00                              | Diabetes                         | Morning glucose                                                                                               |
| 8  | Insulin lispro sliding scale, 6 times daily                     | Diabetes                         | Glucose before every dose                                                                                     |
| 9  | Diltiazem ER 120 mg daily, restarted day 3                      | Heart rate control               | Raises tacrolimus level. Heart rate, blood pressure.                                                          |
| 10 | Sertraline 100 mg daily                                         | Depression                       | QTc, sodium                                                                                                   |
| 11 | Ondansetron 4 mg IV every 8 hours as needed                     | Nausea                           | QTc                                                                                                           |
| 12 | Heparin 5,000 units under the skin every 8 hours                | Prevents clots                   | Platelets, bleeding, drain output                                                                             |
| 13 | Piperacillin-tazobactam 3.375 g IV every 8 hours, started 11:00 | Fever                            | Penicillin allergy flag. Dose depends on kidney function. Cultures.                                           |
| 14 | Pantoprazole 40 mg daily                                        | Stomach protection               | Magnesium                                                                                                     |
| 15 | Magnesium oxide 400 mg twice daily                              | Low magnesium from tacrolimus    | Magnesium level, diarrhea                                                                                     |
| 16 | Aspirin 81 mg daily                                             | Protects the liver artery        | Platelets, bleeding                                                                                           |
| 17 | Amlodipine 5 mg daily                                           | Blood pressure                   | Blood pressure, leg swelling                                                                                  |
| 18 | Oxycodone 5 mg every 6 hours as needed                          | Pain                             | Sedation, breathing rate, bowel movements                                                                     |
| 19 | Acetaminophen 650 mg every 6 hours as needed, 2 g daily limit   | Pain, fever                      | Running daily total                                                                                           |
| 20 | Senna-docusate twice daily                                      | Constipation                     | Stool count                                                                                                   |
| 21 | Lactulose 30 mL three times daily                               | Carried over from admission list | No longer needed. Causes diarrhea and fluid loss.                                                             |
| 22 | Rifaximin 550 mg twice daily                                    | Carried over from admission list | No longer needed                                                                                              |

### On hold or unresolved

| Drug     | Status in the EHR       | Actual status            |
|----------|-------------------------|--------------------------|
| Apixaban | On hold since admission | Restart date not decided |

| Drug           | Status in the EHR                   | Actual status                             |
|----------------|-------------------------------------|-------------------------------------------|
| Furosemide     | On hold since day 2                 | Weight up 1.3 kg since yesterday          |
| Warfarin       | Listed as an active home medication | Replaced by apixaban before August 2026   |
| Metformin      | Listed as an active home medication | Stopped in 2025                           |
| Spironolactone | Listed as an active home medication | Stopped September 2026 for high potassium |
| Ibuprofen      | Not listed anywhere                 | Taken a few times a week at home          |

**Interaction groups the team has to track**

- **Raise the tacrolimus level:** fluconazole, diltiazem
- **Raise potassium:** tacrolimus, trimethoprim-sulfamethoxazole, heparin
- **Stress the kidneys:** tacrolimus, piperacillin-tazobactam, and ibuprofen at home
- **Lengthen the QT interval:** fluconazole, sertraline, ondansetron, tacrolimus
- **Raise glucose:** prednisone, tacrolimus
- **Lower white cells:** mycophenolate, valganciclovir, trimethoprim-sulfamethoxazole

## Medication administration record, last 24 hours

Nurses recorded 32 medication doses in 24 hours, and five of the entries show a problem that no one has acted on. The record is a separate screen from the orders and from the notes.

| Date, time    | Medication                                                                                                                                                       | Status                   | Comment recorded                                       |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------|
| Sep 28, 13:10 | Lispro 8 units                                                                                                                                                   | Given                    | Glucose 301                                            |
| Sep 28, 14:00 | Heparin 5,000 units                                                                                                                                              | Given                    |                                                        |
| Sep 28, 17:05 | Lispro 6 units                                                                                                                                                   | Given                    | Glucose 284                                            |
| Sep 28, 21:00 | Mycophenolate 1,000 mg                                                                                                                                           | <b>Not given</b>         | Patient refused, nausea. Prescriber not told.          |
| Sep 28, 21:00 | Tacrolimus 4 mg                                                                                                                                                  | <b>Given late, 22:40</b> | Dose missing from unit cabinet. Sent up from pharmacy. |
| Sep 28, 21:10 | Glargine 18 units, lispro 4 units                                                                                                                                | Given                    | Glucose 244                                            |
| Sep 28, 21:15 | Lactulose 30 mL                                                                                                                                                  | Given                    |                                                        |
| Sep 28, 22:00 | Acetaminophen 650 mg                                                                                                                                             | Given                    | Pain 4 out of 10                                       |
| Sep 28, 22:00 | Heparin 5,000 units                                                                                                                                              | Given                    |                                                        |
| Sep 29, 01:15 | Oxycodone 5 mg                                                                                                                                                   | Given                    | Pain reassessment not recorded                         |
| Sep 29, 02:05 | Lispro 2 units                                                                                                                                                   | Given                    | Glucose 198                                            |
| Sep 29, 04:00 | Acetaminophen 650 mg                                                                                                                                             | Given                    | Pain 3 out of 10                                       |
| Sep 29, 06:00 | Heparin 5,000 units, lispro 4 units                                                                                                                              | Given                    | Glucose 247                                            |
| Sep 29, 08:10 | Lispro 6 units                                                                                                                                                   | Given                    | Glucose 268                                            |
| Sep 29, 08:40 | Ondansetron 4 mg IV                                                                                                                                              | Given                    | Nausea. QT alert overridden.                           |
| Sep 29, 09:00 | Tacrolimus 4 mg                                                                                                                                                  | <b>Given</b>             | Level not back. No hold order.                         |
| Sep 29, 09:10 | Mycophenolate, prednisone, valganciclovir, trimethoprim-sulfamethoxazole, fluconazole, diltiazem, sertraline, pantoprazole, magnesium oxide, aspirin, amlodipine | Given                    | 11 drugs scanned in 6 minutes                          |
| Sep 29, 09:15 | Lactulose 30 mL, senna-docusate                                                                                                                                  | <b>Given</b>             | Loose stool at 05:00                                   |
| Sep 29, 10:40 | Acetaminophen 650 mg                                                                                                                                             | Given                    | Temperature 38.1 C                                     |
| Sep 29, 11:20 | Piperacillin-tazobactam 3.375 g IV                                                                                                                               | Started                  | Allergy alert overridden by physician                  |
| Sep 29, 12:30 | Lispro 8 units                                                                                                                                                   | Due                      | Glucose 312                                            |

### Problems visible in this record

- 1 The evening tacrolimus dose was 100 minutes late. Today's level was drawn 9 hours 40 minutes after it, not the standard 12 hours.
- 2 One mycophenolate dose was missed, and the prescriber was not told.
- 3 This morning's tacrolimus was given before the level was known, on a patient with a new tremor.
- 4 Two laxatives were given to a patient with loose stools and falling urine output.
- 5 Acetaminophen has reached 1,950 mg of the 2,000 mg daily limit, with a fever still present.

## Physician progress note, post-op day 6

**Date:** September 29, 2026, 09:15. **Author:** Liam Chen, MD. **Service:** Transplant surgery.

The full note prints at 11 pages. About 2 pages are new today. The rest is imported labs, medication lists, and text carried from earlier notes. Lines carried forward unchanged from the September 22 admission note are marked (*carried forward*).

**Overnight events:** None reported.

**Subjective:** Slept poorly. Mild nausea after breakfast. Incision pain 3 out of 10. Walking in the hallway twice daily. Passing gas, one loose stool.

### Objective

- Vitals at 08:00: temperature 37.4 C, heart rate 92, blood pressure 148/88, oxygen 96% on room air
- General: awake, sitting in chair
- Mental status: oriented to person and place, slow to answer, mild hand flapping (*carried forward*)
- Skin and eyes: yellow (*carried forward*)
- Heart: irregular rhythm, no murmur
- Lungs: clear, reduced at right base
- Abdomen: distended with fluid, not tender (*carried forward*)
- Abdomen: incision clean and dry, staples intact. One drain with thin pink fluid.
- Legs: swelling to mid-shin
- Labs: day 6 results pending at time of note

### Assessment and plan

- 1 **Liver transplant, day 6.** Graft function improving. Liver numbers and bilirubin falling daily. Ultrasound yesterday normal. Continue tacrolimus 4 mg twice daily, target level 8 to 12. Follow up today's level. Continue mycophenolate and prednisone 20 mg.
- 2 **Infection prevention.** Continue valganciclovir and trimethoprim-sulfamethoxazole. Fluconazole day 3 of 14 for yeast in drain fluid.
- 3 **Kidney disease.** Creatinine 1.6 yesterday, at baseline. Avoid drugs that harm the kidneys. Kidney team following.
- 4 **Diabetes with steroid effect.** Glucose 198 to 301 over 24 hours. Endocrinology adjusting insulin.
- 5 **Atrial fibrillation.** Rate controlled on diltiazem. Apixaban on hold. Restart when surgically safe.
- 6 **Hypertension.** Blood pressure above goal. Continue amlodipine. Consider increase tomorrow.
- 7 **Hepatic encephalopathy, grade 2.** Continue lactulose and rifaximin. Adjust lactulose to 3 soft stools daily. (*carried forward*)
- 8 **Ascites.** Continue diuretics. Drain fluid as needed for comfort. (*carried forward*)
- 9 **Depression.** Continue sertraline.
- 10 **Nutrition and mobility.** Regular diabetic diet. Physical therapy daily.
- 11 **Discharge.** Target October 2, day 9. Case management aware.

**Addendum, 11:05.** Temperature 38.1 C at 10:30, heart rate 104. Blood and urine cultures sent. Started piperacillin-tazobactam. Penicillin allergy noted; patient tolerated cefazolin in surgery. Will follow cultures.

**Addendum, 11:06.** Alerts acknowledged: allergy, QT, duplicate insulin, kidney dosing, drug interaction.

**What this note does not say**

- It does not mention that fluconazole and diltiazem raise tacrolimus levels.
- It does not mention the potassium of 5.0 or the falling urine output.
- It says "continue diuretics," but furosemide has been on hold since day 2.
- It does not record the kidney team's advice from September 28.

## Consult notes

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Eight consulting services wrote notes between September 26 and 29. Three of them warned about the tacrolimus level. Each note sits in a different tab of the chart, and none is referenced in the surgical progress note.

**Infectious disease.** September 27, 15:30. Author: Samuel Reyes, MD.

- Reason: yeast (*Candida albicans*) in surgical drain fluid, post-op day 4.
- Assessment: likely early abdominal yeast infection in an immunosuppressed patient. No fever at this time.
- Recommend fluconazole 200 mg daily for 14 days, adjusted for kidney function.
- **Fluconazole will raise tacrolimus levels. Suggest reducing the tacrolimus dose and re-checking the level in 48 to 72 hours.**
- Cytomegalovirus: high-risk pairing. Valganciclovir for 6 months. Weekly blood test for the virus.
- Penicillin allergy is low risk. He tolerated cefazolin. Recommend removing the allergy label.
- Follow-up: will see again if fever develops.

**Endocrinology.** September 28, 14:20. Author: Grace Lindqvist, MD.

- Reason: glucose 218 to 301 on prednisone and tacrolimus.
- Assessment: type 2 diabetes with steroid-induced high glucose. Both prednisone and tacrolimus raise glucose.
- Increase glargine from 14 to 18 units nightly. Continue lispro sliding scale.
- Check glucose 6 times daily, including 02:00.
- Insulin needs will fall as prednisone tapers. Reduce doses with each taper step to avoid low glucose.
- Diabetes education for patient and daughter before discharge.

**Nephrology.** September 28, 16:40. Author: Helen Park, MD.

- Reason: chronic kidney disease in a new transplant patient on tacrolimus.
- Findings: creatinine 1.6, potassium 5.0 and rising for 3 days, urine output falling, weight up.
- Assessment: at high risk of acute kidney injury. Tacrolimus level has doubled in 48 hours since fluconazole and diltiazem were started.
- **If creatinine is above 2.0 or tacrolimus is above 12 tomorrow, hold the next tacrolimus dose and call nephrology.**
- Repeat potassium and creatinine at 06:00 and 14:00 tomorrow.
- Consider stopping trimethoprim-sulfamethoxazole if potassium passes 5.5.
- No ibuprofen or similar drugs. Low-potassium diet.
- Outside kidney clinic records still not received. Baseline before 2026 unknown.

**Transplant pharmacy.** September 29, 09:50. Author: Dana Patel, PharmD.

- Reason: medication reconciliation.
- Three home medication lists disagree. See admission note table.
- Daughter confirms by phone: no warfarin, no metformin, no spironolactone. Takes ibuprofen a few times a week.
- Recommend stopping lactulose and rifaximin. No longer needed after transplant.
- Recommend removing warfarin, metformin, and spironolactone from the home list.
- Valganciclovir dose to be re-checked against today's creatinine.
- Tacrolimus interaction review: waiting for today's level.

- Discharge: tacrolimus and valganciclovir need prior authorization from the insurer. Request sent September 28, no reply yet.

**Cardiology.** September 26, 11:10. Author: Omar Haddad, MD.

- Reason: atrial fibrillation with heart rate 150 on day 2. Heart stent in 2023.
- Assessment: fast atrial fibrillation triggered by surgery, blood loss, and low magnesium.
- Continue diltiazem. **Diltiazem raises tacrolimus levels. Transplant team to adjust the dose.**
- Continue aspirin for the stent. Do not stop it.
- Blood thinner: he has atrial fibrillation and a past portal vein clot, and he bled on day 1. Restart by day 7 if drain output stays low. Decision belongs to the surgeon.
- QTc 460 ms. Avoid adding QT-lengthening drugs. Keep potassium above 4.0 and magnesium above 2.0.
- Haloperidol given last night. Do not repeat.

**Nutrition.** September 27, 13:45. Author: Leah Goldberg, RD.

- Assessment: severe malnutrition with muscle loss. Albumin 2.4. Eating about 40% of meals.
- Needs 2,200 calories and 110 g protein daily.
- Ordered high-protein supplement drinks three times daily. Each contains 400 mg potassium.
- A low-potassium diet was ordered by nephrology on September 28. The supplement order was not changed.
- If intake stays below 50% for 2 more days, recommend a feeding tube.

**Social work.** September 28, 10:30. Author: Denise Carter, LCSW.

- Lives alone. Daughter works full time and can take 2 weeks of leave.
- Transplant protocol requires a caregiver around the clock for the first 4 weeks. No plan for weeks 3 and 4.
- Missed 2 of his last 6 clinic visits for lack of transport.
- Depression screening score 11, moderate. He says he is "scared of messing up the pills."
- Daughter asked whether "the cancer is gone now." No one has given her the pathology result.
- Recommend family meeting before any discharge date is set.

**Physical therapy.** September 29, 10:15. Author: Ben Aldridge, PT.

- Walked 60 meters with a walker and one person helping.
- Heart rate rose to 124 and oxygen fell to 91% while walking. Stopped early.
- High fall risk. Cannot climb stairs. His home has 6 steps at the entrance.
- Recommend discharge to an inpatient rehabilitation facility, not home.
- This conflicts with the case management plan for discharge home on October 2.

## Nursing notes, post-op day 6

The night nurse recorded shaky hands and low urine output at 05:30, seven hours before the labs resulted. Both findings are in free-text nursing notes that physicians rarely open.

**Night shift note.** September 29, 06:30. Author: Karen Osei, RN.

- Slept in short stretches. Pain 2 to 4 out of 10, oxycodone given once at 01:15.
- Glucose 198 at 02:00, 2 units lispro given. Glucose 247 at 06:00, 4 units given.
- **05:30: patient says his hands "feel shaky" holding a cup. Thought to be poor sleep.**
- **Urine output 210 mL this shift, darker than yesterday.** Surgery resident paged at 05:50. No callback recorded.
- One loose stool after lactulose.
- Labs drawn at 05:40. Lab called at 07:10: sample clotted, redraw needed. Redrawn at 08:20, after shift change.
- Weight 94.2 kg, up 1.3 kg from yesterday.
- Handoff to day shift: surgery "might change the tacrolimus." No new order seen.

**Day shift note.** September 29, 12:45. Author: Maya Torres, RN.

- 07:30: full assessment done. Alert and oriented to person, place, time, and situation. No hand flapping.
- 09:00: tacrolimus 4 mg given as ordered. No hold order in the chart. Morning level not back.
- 09:10: remaining morning medications given, 11 in total.
- 10:30: temperature 38.1 C, heart rate 104. Dr. Chen told at 10:35.
- 10:50: blood cultures from 2 sites and urine culture sent.
- 11:20: first dose of piperacillin-tazobactam started. Allergy alert overridden by physician. Watched 30 minutes, no reaction.
- 12:00: vitals repeated. Heart rate 108, irregular. Blood pressure 156/92.
- 12:30: glucose 312. Lispro 8 units due.
- Walked 60 meters with physical therapy at 10:00. Tired afterward.
- Daughter phoned twice asking about discharge date and the fever.
- Patient anxious about fever. Reassured.

### Required assessments this shift

| Assessment                                | Result                             | Next due  |
|-------------------------------------------|------------------------------------|-----------|
| Fall risk score                           | 45, moderate risk                  | 16:00     |
| Skin and pressure injury score            | 17, mild risk                      | 19:00     |
| Pain reassessment after oxycodone         | Not done at 02:15, flagged overdue | Overdue   |
| Surgical drain: amount and color          | 60 mL, thin pink                   | 16:00     |
| Intake and output                         | 900 mL in, 210 mL out              | 16:00     |
| Neurological check                        | Normal at 07:30                    | 15:30     |
| Incision check                            | Clean, dry                         | 19:00     |
| Central line dressing                     | Changed September 27               | October 4 |
| Patient education: transplant medications | Session 2 of 4 complete            | Tomorrow  |

The same vitals, weight, and drain output are also entered in the transplant module and the intake and output screen.

## Monitoring schedule

On day 6 the care team has 15 separate things to check on this patient, adding up to more than 60 chart entries in 24 hours. Each entry can trigger alerts, and most of them feed a dose decision.

| What is checked             | How often                                     | Why                                                                    | Action trigger                                           |
|-----------------------------|-----------------------------------------------|------------------------------------------------------------------------|----------------------------------------------------------|
| Tacrolimus blood level      | Daily, before the 09:00 dose                  | Too low risks rejection. Too high harms kidneys and nerves.            | Outside 8 to 12: change dose. Above 15: hold and call.   |
| Creatinine and urine output | Labs daily; urine every 4 hours               | Kidney disease plus three kidney-stressing drugs                       | Creatinine above 2.0, or urine under 30 mL per hour      |
| Potassium                   | Daily; twice daily if above 5.0               | Tacrolimus, trimethoprim-sulfamethoxazole, and heparin all raise it    | Above 5.5: treat and review drugs. Above 6.0: emergency. |
| Magnesium                   | Daily                                         | Tacrolimus lowers it. Low magnesium worsens heart rhythm.              | Below 1.5: replace                                       |
| Liver numbers and bilirubin | Daily                                         | Earliest sign of rejection or a blocked artery                         | Any rise after a steady fall                             |
| INR                         | Daily                                         | Shows whether the new liver is working                                 | Rising INR                                               |
| Glucose                     | 6 times daily                                 | Steroids and tacrolimus raise it. Insulin can drop it.                 | Below 70 or above 300                                    |
| Vital signs                 | Every 4 hours                                 | Fever may be the only sign of infection in an immunosuppressed patient | Temperature 38.0 C or higher. Heart rate above 110.      |
| Heart rhythm and QTc        | Continuous monitor; ECG every 2 days          | Atrial fibrillation plus 4 QT-lengthening drugs                        | QTc above 500 ms. Heart rate above 130.                  |
| Surgical drain output       | Every 4 hours                                 | Blood means bleeding. Green or brown means a bile leak.                | Color change, or more than 100 mL per hour               |
| White cells and platelets   | Daily                                         | Infection, and drug effects on bone marrow                             | White cells below 3 or above 15. Platelets below 50.     |
| Mental status and tremor    | Every shift                                   | Tacrolimus toxicity shows as tremor, headache, confusion               | Any new tremor or confusion                              |
| Weight and fluid balance    | Daily weight; intake and output every 4 hours | Fluid overload and kidney function                                     | Gain over 1 kg in a day                                  |
| Cultures and virus tests    | When fever occurs; virus test weekly          | High-risk cytomegalovirus pairing. Yeast in drain.                     | Any positive result                                      |
| Drug interaction review     | With every new order                          | 6 overlapping interaction groups                                       | Any new drug that changes tacrolimus levels              |

### Decisions that depend on these checks every day

- 1 Tacrolimus dose for tonight and tomorrow
- 2 Insulin doses at each meal and at bedtime
- 3 Whether to restart the blood thinner
- 4 Whether to restart the fluid pill
- 5 Kidney dose adjustments for valganciclovir and the antibiotic
- 6 Potassium and magnesium replacement or restriction
- 7 Whether the fever means infection, rejection, or a drug reaction

**Triggers already met by 12:45 on September 29, before the labs resulted**

- New tremor reported at 05:30
- Urine output 26 mL per hour over 8 hours
- Weight gain of 1.3 kg in a day
- Temperature 38.1 C
- A new drug that raises tacrolimus, with no dose change

## Discharge planning note

**Date:** September 29, 2026, 12:50. **Author:** Rosa Alvarez, RN, case manager.

Discharge is targeted for October 2, and none of the 9 required items is complete. Each item is tracked in a different screen or by phone.

| Item                                               | Where it is tracked          | Status                                           |
|----------------------------------------------------|------------------------------|--------------------------------------------------|
| Transplant clinic visits, twice weekly for 4 weeks | Outpatient scheduling system | First visit booked October 5. Others not booked. |
| Lab draws, twice weekly                            | Paper standing order, faxed  | Local lab in Carver County not yet confirmed     |
| Tacrolimus and valganciclovir supply               | Specialty pharmacy portal    | Prior authorization pending since September 28   |
| Other discharge prescriptions                      | EHR discharge module         | Blocked: medication reconciliation incomplete    |
| Home health nursing                                | Phone and fax                | Agency accepted, start date open                 |
| Transport, 90 miles each way                       | Phone                        | Daughter can drive Mondays only                  |
| Diabetes teaching                                  | Education module             | Not started                                      |
| Transplant medication teaching                     | Education module             | Session 2 of 4 complete                          |
| Follow-up with outside kidney doctor               | Phone                        | Clinic has not returned 2 calls                  |

**Barriers:** distance from the hospital, one caregiver with a full-time job, 14 discharge medications, new insulin regimen, and a blood thinner decision still open.